A Robust Numerical Algorithm for simulating the interaction of multiphase and multimaterial fluid(s) with flexible solid(s) *

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ABSTRACT: A new Eulerian method is presented for simulating complex three dimensional and three dimensional axisymmetric flows consisting of any number of materials, to include compressible fluids, incompressible fluids, ice, melt, vapor, and solids undergoing plastic deformation. The interfaces between materials is modeled and discretized as sharp. The new method is robust with respect to a wide range of material properties, and large interface deformation. The new method is scalable on MIMD (Multi Instruction Distributed memory using MPI) and MISD (Multi Instructuion Shared memory using OpenMP)

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computer systems. The new method is also scalable with respect to dynamic adaptive mesh refinement.

1 Introduction

The ability to simulate multiphase and multimaterial high and low speed flows, to include the interaction of fluid(s) and elastic material(s), has a number of applications: bubble growth and collapse interacting with flexible surfaces[46, 47, 42], wetting dynamics of liquid droplets on deformable membranes[29], design of breathable masks[12], underwater explosions near structures[26, 49, 8, 15, 18, 43, 34], hydroelastic applications[28], vibration of pipes due to internal boiling flow[2], ultrasonic leak detection in nuclear power plants[10], and the measurement of the tensile strength properties of colliding and deforming structures[11].

We are not aware of previously reported research on the robust numerical simulation of the interaction of multiphase flows with elastic materials, to include incompressible materials, weakly compressible materials, and/or strongly compressible materials. The research that we report here is a continuation of research that started with the following article by Jemison et al[20]. In Jemison et al's research, a novel adaptive, parallel, asymptotically preserving[23, 5, 6, 35], moment-of-fluid method was presented for simulating two phase flows to include incompressible, low speed, and high speed flows. The method in Jemison et al was a collocated generalization of the projection method for incompressible flows[19] in order to simulate both compressible (with shock waves) and incompressible flows.

In the present work, we have improved upon [20] in the following important ways:

1. the present method is now a staggered grid method instead of colocated. See Figure **??**. The following research for incompressible flows underscore the importance of a staggered grid[25, 37, 36, 17] for projection methods. We also report recent work by Park and Son[31] in which they developed a staggered grid method for compressible flows in which the acoustic time step constraint was avoided.

2. the present method represents complex deforming interfaces using the Continuous Moment-of-Fluid method[48] (CMOF), instead of the Moment-of-Fluid (MOF) method[21]. The CMOF method[48] removes a "checkerboard instability" that appears for surface tension driven flows.

3. We have developed a conservative, staggered grid, algorithm for solving the equations associated with standard elastic materials and materials undergoing plastic deformation. Our new algorithm is an improvement on the algorithms reported in [37, 36, 31] so that we can simulate both low and high speed fluid structure interaction problems. Our new method can simulate incompressible elastic materials or compressible elastic materials with shock waves.

4. We have developed an exactly mass and volume preserving algorithm for simulating contact line dynamics of drops and bubbles along flexible surfaces.

In Table 1, we compare and contrast our new method with the current state of the art.

1.1 Motivation for enhancing our asymptotically preserving algorithm from collocated grid to staggered grid

In this section, we just focus on single material inviscid compressible gas dynamics with a general equation of state and periodic boundary conditions. The derivations here are applicable to the general multimaterial case. The compressible gas dynamic equations[1, 6, 35, 9] are:

$$\rho_t + \nabla \cdot (\rho \vec{u}) = 0 \tag{1}$$

$$(\rho \vec{u})_t + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p\mathbb{I}) = 0 \tag{2}$$

$$(\rho E)_t + \nabla \cdot (\rho \vec{u} E + p\vec{u}) = 0 \tag{3}$$

$$E = e + \frac{1}{2}\vec{u} \cdot \vec{u} \tag{4}$$

$$p = p(\rho, e) \qquad c^2 = p_\rho + \frac{p_e p}{\rho^2} \qquad e = c_v T \tag{5}$$

Following the "asymptotically preserving" analysis in these articles, [1, 6, 35, 9], it can be shown that the Directionally split Cell Integrated Semi-Lagrangian (CISL) algorithm given by Algorithm 1 approaches a discretization consistent with the incompressible flow equations in the zero Mach number limit.

The advection commands in Algorithm1 are discretized using the directionally split cell integrated semi-Lagrangian advection algorithm with piecewise constant reconstruction. The relation between the variables' control volumes and the computational mesh are shown in Figure **??**. The piecewise constant representation guarantees that no new extrama are generated for $\vec{u}^*$ and E^*, the updated density ρ^{n+1} is positive, and the discrete mass, momentum, and energy are conserved.

For incompressible flows, it has already been shown that the staggered grid discretization [25] of steps 9 and 10 in algorithm 1 results in a provably stable and convergent method.

For the compressible case, if we consider a simplified problem, in which the staggered grid discretization of steps 9 and 10 in algorithm 1 is applied to the wave equation:

$$\rho \vec{u}_t = -\nabla p \qquad \rho = \rho(\vec{x}) \ \rho \text{ is positive} \tag{6}$$

Author(s)	incompressible fluids	compressible fluids	multiphase flow	multimaterial flow	shock waves	Coupling Method	Interface Representation	Complex topology
Wardlaw and Luton [43]	No	Yes	No	Yes	Yes	partitioned	ALE (fluid interfaces) Lagrangian (structure)	Yes
Tran and Udaykumar [39]	No	Yes	No	Yes	Yes	Explicit	Particle Level Set	Yes
Robinson-Mosher et al. [33]	Yes	No	No	No	No	Monolithic	Connected Markers	No
Gretarsson et al. [16]	No	Yes	No	No	Yes	Monolithic	one dimension	No
Sugiyama et al. [37]	Yes	No	No	No	No	pressure and velocity	Volume of fluid	yes
Banks et al. [3]	Yes	No	No	No	No	Partitioned	Body fitted	No
Patkar et al. [32]	No	Yes	No	No	Yes	Monolithic	Connected Markers	No
Banks et al. [4]	Yes	No	No	No	No	Partitioned	Body fitted	No
Shin et al. [36]	Yes	No	Yes	Yes	No	pressure and velocity	level contour reconstruction	Yes
Mokbel et al. [29]	Yes	No	No	Yes	No	Monolithic	Hybrid Diffuse and ALE	No
Kloppe and Aland [22]	Yes	No	No	Yes	No	Monolithic	Hybrid Diffuse and ALE	No
Gao et al. [14]	No	Yes	No	Yes	Yes	explicit moving mesh ALE	moving mesh ALE	Yes
Ferrari et al. [13]	No	Yes	Yes	Yes	Yes	explicit non-partitioned	Diffuse interface	Yes
Wilfong et al. [44]	No	Yes	Yes	Yes	Yes	explicit non-partitioned	Diffuse interface	Yes
Park and Son [31]	Yes	Yes	Yes	Yes	No	pressure and velocity	Coupled level set volume of fluid	Yes
Present Article, Estebe et al	Yes	Yes	Yes	Yes	Yes	pressure and velocity	Continuous Moment of Fluid	Yes

Table 1: Recent numerical methods with fluid interaction with deformable structures. Legend: ALE (Arbitrary Lagrangian Eulerian), "pressure and velocity coupling" refers to the asymptotically preserving methods which reduce to the projection method in the zero Mach number limit[31, 37, 36], "Partitioned coupling" refers to alternating solving the fluid equations with elastic material boundary conditions and then solving the elastic material equations with fluid boundary conditions.

Algorithm 1 Asymptotically Preserving time discretization for gas dynamics

$k \leftarrow 0$
$\vec{u}^{n+1,(k)} \leftarrow \vec{u}^n$
while $k \leq 1$ **do**
  $\rho^{n+1,(k+1)} \leftarrow \rho^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho \vec{u}^{n+1,(k)}) dt$
  $\rho^{n+1,(k+1)} \vec{u}^{*,(k+1)} \leftarrow (\rho \vec{u})^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho \vec{u} \otimes \vec{u}^{n+1,(k)}) dt$
  $\rho^{n+1,(k+1)} E^{*,(k+1)} \leftarrow (\rho E)^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho E \vec{u}^{n+1,(k)}) dt$
  $p^{*,(k+1)} \leftarrow p(\rho^{n+1,(k+1)}, e^{*,(k+1)})$
  $c^{*,(k+1)} \leftarrow c(\rho^{n+1,(k+1)}, e^{*,(k+1)})$
  $\vec{u}^{n+1,(k+1)} \leftarrow \vec{u}^{*,(k+1)} - \Delta t \frac{\nabla p^{n+1,(k+1)}}{\rho^{n+1,(k+1)}}$
  $p^{n+1,(k+1)} \leftarrow p^{*,(k+1)} - \Delta t \rho^{n+1,(k+1)} (c^{*,(k+1)})^2 \nabla \cdot \vec{u}^{n+1,(k+1)}$
  $E^{n+1,(k+1)} \leftarrow E^{*,(k+1)} = -\Delta t \nabla \cdot (\vec{u} p)^{n+1,(k+1)}$
  $k \leftarrow k + 1$
end while

$$\frac{1}{\rho c^2} p_t = -\nabla \cdot \vec{u} \qquad c = c(\vec{x}) \text{ } c \text{ is positive} \tag{7}$$

then the staggered grid approach guarantees unconditional stability for any fixed

$$\frac{\Delta t}{\Delta x} \tag{8}$$

1.2 Lax diffusion for elastic materials

In this section, we just focus on single material compressible elastic fluid dynamics with a general equation of state and periodic boundary conditions. The derivations here are applicable to the general multimaterial case.
The compressible elastic fluid dynamic equations[39, 27, 7] are:

$$\rho_t + \nabla \cdot (\rho \vec{u}) = 0 \tag{9}$$

$$(\rho \vec{u})_t + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p \mathbb{I} - s) = 0 \tag{10}$$

$$(\rho E)_t + \nabla \cdot (\rho \vec{u} E + p \vec{u} - \vec{u} \cdot s) = 0 \tag{11}$$

$$E = e + \frac{1}{2} \vec{u} \cdot \vec{u} \tag{12}$$

$$(\rho s)_t + \nabla \cdot (\rho \vec{u} s) = \rho(\Omega s - s\Omega + 2G(\bar{D} - D^p)) \tag{13}$$

$$D = \frac{\nabla \vec{u} + (\nabla \vec{u})^T}{2} \tag{14}$$

$$\Omega_{ij} = \frac{1}{2}(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}) \tag{15}$$

$$p = p(\rho, e) \qquad c^2 = p_\rho + \frac{p_e p}{\rho^2} \qquad e = c_v T \tag{16}$$

D^p is the plastic strain rate and is defined below (38).
The elastic tensor s is discretized on a staggered grid as illustrated in Figure **??** and as recommended by [37, 36, 31].
We discretize the elastic term, $\nabla \cdot s$ in (10), and the $\nabla \cdot (\vec{u}p)$ term in (11) using central differencing for which we have found to cause problems in elastic materials for realistic values of the shear modulus G. In order to overcome this difficulty, we add Lax Friedrichs Diffusion[24] to both the momentum equation (10),

$$\nabla \cdot (2\mu_{\text{Lax}} D), \tag{17}$$

$$\mu_{\text{Lax}} = \frac{\rho_{\text{max}} \Delta x^2}{2\text{DIM}\Delta t} \qquad D = \frac{\nabla \vec{u} + (\nabla \vec{u})^T}{2} \qquad \text{DIM} = 2 \text{ or } 3, \tag{18}$$

and the energy equation (11),

$$\nabla \cdot (k_{\text{Lax}} \nabla T), \tag{19}$$

$$k_{\text{Lax}} = \frac{\rho_{\text{max}} c_v \Delta x^2}{2\text{DIM}\Delta t}. \tag{20}$$

So, an outline of our asymptotically preserving temporal discretization of non-isothermal gas dynamics, including Lax-Friedrichs smoothing, is given by Algorithm 2.

Remarks:

1. In [24], the artificial diffusion term(s) are discretized explicitly in time, but in our implementation, diffusion is discretized in time semi-implicitly; see steps 9 and 10 in Algorithm 2.

2. In [24], it is proved for general scalar hyperbolic problems that the Lax method converges if the CFL condition is met. For systems of hyperbolic conservation laws, Lax conjectures that the method converges, but does not offer a proof. We have a similar conjecture, that Lax artificial diffusion guarantees convergence; our conjecture is stronger. We conjecture that adding Lax artificial diffusion to our method guarantees convergence so long as the "velocity" CFL condition and "shear modulus" CFL conditions are met (**??**), but sound speed restrictions on the time step are unnecessary.

Algorithm 2 Asymptotically Preserving time discretization for an elastic material

$k \leftarrow 0$
$\vec{u}^{n+1,(k)} \leftarrow \vec{u}^n$
while $k \leq 1$ **do**
  $\rho^{n+1,(k+1)} \leftarrow \rho^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho \vec{u}^{n+1,(k)}) dt$
  $\rho^{n+1,(k+1)} \vec{u}^{*,(k+1)} \leftarrow (\rho \vec{u})^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho \vec{u} \otimes \vec{u}^{n+1,(k)}) dt$
  $\rho^{n+1,(k+1)} E^{*,(k+1)} \leftarrow (\rho E)^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho E \vec{u}^{n+1,(k)}) dt$
  $\rho^{n+1,(k+1)} s^{*,(k+1)} \leftarrow (\rho s)^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho s \vec{u}^{n+1,(k)}) dt$
  $s^{n+1,(k+1)} \leftarrow s^{*,(k+1)} + \Delta t(\Omega^{*,(k+1)} s^{*,(k+1)} - s^{*,(k+1)} \Omega^{*,(k+1)} + 2G(\bar{D}^{*,(k+1)} - (D^p)^{*,(k+1)}))$
  $\vec{u}^{**,(k+1)} \leftarrow \vec{u}^{*,(k+1)} + \Delta t \frac{\nabla \cdot (2(\mu + \mu_{\text{Lax}}) D^{**,(k+1)})}{\rho^{n+1,(k+1)}}$
  $T^{**,(k+1)} \leftarrow T^{*,(k+1)} + \Delta t \frac{\nabla \cdot ((k + k_{\text{Lax}}) \nabla T^{**,(k+1)})}{\rho^{n+1,(k+1)} c_v}$
  $\vec{u}^{***,(k+1)} \leftarrow \vec{u}^{**,(k+1)} + \Delta t \frac{\nabla \cdot s^{*,(k+1)}}{\rho^{n+1,(k+1)}}$
  $T^{***,(k+1)} \leftarrow T^{**,(k+1)} + \Delta t \frac{\nabla \vec{u}^{***,(k+1)} : s^{*,(k+1)}}{\rho^{n+1,(k+1)} c_v}$
  $p^{*,(k+1)} \leftarrow p\big(\rho^{n+1,(k+1)}, e^{***,(k+1)}\big)$
  $c^{*,(k+1)} \leftarrow c\big(\rho^{n+1,(k+1)}, e^{***,(k+1)}\big)$
  $\vec{u}^{n+1,(k+1)} \leftarrow \vec{u}^{***,(k+1)} - \Delta t \frac{\nabla p^{n+1,(k+1)}}{\rho^{n+1,(k+1)}}$
  $p^{n+1,(k+1)} \leftarrow p^{*,(k+1)} - \Delta t \rho^{n+1,(k+1)} \big(c^{*,(k+1)}\big)^2 \nabla \cdot \vec{u}^{n+1,(k+1)}$
  $E^{n+1,(k+1)} \leftarrow E^{***,(k+1)} = -\Delta t \nabla \cdot (\vec{u} p)^{n+1,(k+1)}$
  $k \leftarrow k + 1$
end while

3. We have done extensive testing of our numerical method and have found that Lax artificial diffusion to be necessary only for stiff elastic materials. We have found that adding Lax diffusion to liquid materials would impose excessive damping.

4. We are only aware of proofs for the Godunov method and fractional step Lax-Friedrichs method for isentropic gas dynamics[45]. Neither of these methods are asymptotically preserving.

5. Our numerical algorithm never requires the solution of nonlinear algebraic equations, thereby avoiding robustness problems in which a discrete solution cannot be found or avoiding problems in which the discrete solution is not unique.

2 Equations for Multiphase and Multimaterial flows interacting with elastic materials.

There are M materials and each material $1 \leq m \leq M$ occupies the region,

$$\Omega_m(t) = \{(\vec{x}, t) | \phi_m(\vec{x}, t) \geq 0\} \tag{21}$$

The $\phi_m(\boldsymbol{x}, t)$ are level set functions[38, 30]. The zero level set of such functions demarcate the boundary of material m:

$$\partial\Omega_m(t) = \{(\vec{x}, t)) | \phi_m(\vec{x}, t) = 0\} \tag{22}$$

In the bulk regions of material m we have the following equations for compressible materials:

$$\rho_t + \nabla \cdot (\rho \vec{u}) = 0 \tag{23}$$

$$(\rho \vec{u})_t + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p\mathbb{I} - \tau) = \vec{F}_{body} \tag{24}$$

$$(\rho E)_t + \nabla \cdot ((\rho E + p)\vec{u} - \vec{u} \cdot \tau - k\nabla T) = \vec{u} \cdot \vec{F}_{body} \tag{25}$$

$$(\rho Y)_t + \nabla \cdot (\rho Y \vec{u}) = \nabla \cdot (\rho D \nabla Y) \tag{26}$$

$$\tau = 2\mu \bar{D} \tag{27}$$

$$\bar{D} = D - \frac{\text{Trace}(D)\mathbb{I}}{\text{Trace}(\mathbb{I})} \tag{28}$$

$$D = \frac{\nabla \vec{u} + (\nabla \vec{u})^T}{2} \tag{29}$$

$$\rho E = \rho e_{\text{int}} + \frac{1}{2}\rho|\vec{u}|^2 \tag{30}$$

$$p = p(\rho, e) \quad e = c_v T \tag{31}$$

If the material is a viscoelastic FENE-CR material, then one has the additional equation for the configuration tensor A,

$$(\rho A)_t + \nabla \cdot (\rho \vec{u} A) = \rho(A \cdot (\nabla \vec{u})^T + \nabla \vec{u} \cdot A - \frac{f(R)}{\lambda}(A - \mathbb{I})), \tag{32}$$

and the momentum constitutive law term is modified,

$$\tau = \tau + \frac{\tilde{\eta}_P}{\lambda} f(R) A. \tag{33}$$

If the material is an elastic material undergoing plastic deformation, then one has the following additional equation for the deviatoric stress tensor,

$$(\rho s)_t + \nabla \cdot (\rho \vec{u} s) = \rho(\Omega s - s\Omega + 2G(\bar{D} - D^p)), \tag{34}$$

and the momentum constitutive law term is modified,

$$\tau = \tau + s. \tag{35}$$

If the material is an elastic material undergoing plastic deformation, then the plastic strain equation is solved too,

$$\epsilon_t^p + \vec{u} \cdot \nabla \epsilon^p = \sqrt{\frac{2}{3}}|\Lambda|. \tag{36}$$

$$|\Lambda| = \sqrt{\frac{3}{2}}\sqrt{D^p : D^p} \tag{37}$$

$$D^p = \Lambda N \quad N = \sqrt{3/2}\frac{s}{S_e} \quad (\dot{\epsilon}^p)^2 = \frac{2}{3}\Lambda^2 \tag{38}$$

Ω is the spin tensor,

$$\Omega_{ij} = \frac{1}{2}(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}) \tag{39}$$

The effective stress is defined as,

$$S_e = \sqrt{\frac{3}{2}}\sqrt{s : s} \tag{40}$$

The Johnson-Cook model yield stress is given by,

$$\sigma_y = (A + B(\epsilon^p)^n)(1 + C\ln(\frac{\dot{\epsilon}^p}{\dot{\epsilon}_0}))(1 - \theta^m) \quad \theta = \frac{T - T_0}{T_m - T_0} \tag{41}$$

One form of the Zerilli Armstrong model yield stress is given by,

$$\beta = \beta_0 - \beta_1 \log(\dot{\epsilon}^p / \dot{\epsilon}^p_{ref}) \tag{42}$$

$$\sigma_y = A + B(\epsilon^p)^n e^{-\beta T} \tag{43}$$

The energy contribution due to elasto-plastic deformation is modeled as,

$$\beta \dot{\epsilon}^p S_e, \tag{44}$$

and is added to (25). In order to determine Λ, as it appears in (38), the following "radial return algorithm" is implemented (see [39] for the case when $n = 1$ in (41) or (43)):

1. $\dot{\epsilon}^p = 0$, $k = 0$

2. $S_e = \sqrt{\frac{3}{2}}\sqrt{s : s}$

3. $\sigma_y = \sigma_y(T, \epsilon^p, \dot{\epsilon}^p)$

4. if $S_e - \sigma_y \leq 0$ then $\Lambda = 0$ (i.e. $D^p : D^p = 0$).

5. if $S_e > \sigma_y$ then,

6. Solve the following equation for ϵ^p_{new}:

$$S_e - \sigma_y = 3G(\epsilon^p_{new} - \epsilon^p) + B((\epsilon^p_{new})^n - (\epsilon^p)^n) \tag{45}$$

7.

$$\dot{\epsilon}^p = \frac{\epsilon^p_{new} - \epsilon^p}{\Delta t} \tag{46}$$

8.

$$\Lambda = \sqrt{\frac{3}{2}}\dot{\epsilon}^p \tag{47}$$

$$D^p = \Lambda N \tag{48}$$

9. $k = k + 1$

10. if $k = 1$ then go back to step 3.

The mathematical model at sharp interfaces is given as follows. First, define a "closest point" function:

$$\vec{x}_c(\vec{x}) = \text{argmin}_{\vec{y},m}\, \{||\vec{y}-\vec{x}|| | \phi_m(\vec{y}) = 0\} \tag{49}$$

Taking into account mass transfer, a source material's level set function (e.g. ice that is melting or liquid that is turning into vapor) satisfies

$$(\phi_{m_s})_t + \vec{u}_{m_s}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_s} = -\frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_s}}|\nabla\phi_{m_s}| \tag{50}$$

The destination material's level set function satisfies

$$(\phi_{m_d})_t + \vec{u}_{m_s}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_d} = \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_s}}|\nabla\phi_{m_d}| \tag{51}$$

The above equations are equivalent to the following:

$$(\phi_{m_s})_t + \vec{u}_{m_d}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_s} = -\frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_d}}|\nabla\phi_{m_s}| \tag{52}$$

$$(\phi_{m_d})_t + \vec{u}_{m_d}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_d} = \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_d}}|\nabla\phi_{m_d}| \tag{53}$$

Now, we define the interface normal:

$$\vec{n}_m = \frac{\nabla\phi_m}{||\nabla\phi_m||} \tag{54}$$

The interface curvature is,

$$\kappa_m = \nabla \cdot \vec{n}_m \tag{55}$$

The rate of mass transfer is,

$$\dot{m} = \frac{k_{m_s}\nabla T_{m_s} - k_{m_d}\nabla T_{m_d}}{L} \cdot \vec{n}_{m_d}. \tag{56}$$

The phase change interface temperature condition is,

$$T_{m_s} = T_{m_d} = T_{\text{saturation}}(P_{\text{equilibrium}}) \tag{57}$$

For evaporation, the following rate of mass transfer should match that in (56):

$$\dot{m} = \frac{-\vec{n}_{m_s} \cdot \nabla Y_{m_d}\rho_{m_d} D}{Y^{\Gamma} - 1} \tag{58}$$

Let the tensor σ represent pressure, viscosity, viscoelastic, and elastic contributions to the momentum equations. One has:

$$\sigma = -p\mathbb{I} + \tau. \tag{59}$$

The stress jump conditions are

$$\vec{n}_{m_s} \cdot (-\sigma_{m_s} + \sigma_{m_d}) \cdot \vec{n}_{m_s} = -\gamma_{m_s,m_d}\kappa_{m_s} - \dot{m}^2(\frac{1}{\rho_{m_s}} - \frac{1}{\rho_{m_d}}) \tag{60}$$

$$(\vec{t}_{i,tangent})_{m_s} \cdot (-\sigma_{m_s} + \sigma_{m_d}) \cdot \vec{n}_{m_s} = 0 \quad i = 1, 2 \tag{61}$$

The velocity jump conditions are

$$(\vec{u}_{m_s} - \vec{u}_{m_d}) \cdot \vec{n}_{m_s} = \dot{m}(\frac{1}{\rho_{m_d}} - \frac{1}{\rho_{m_s}}) \tag{62}$$

3 Positioning of discrete variables on the computational grid

Note: for conservation form, velocity on face centroids, off-diagonal configuration tensor variables on edge centroids, the density is stored on a refined mesh.

Remarks:

1. The positioning of discrete variables in our earlier work [20] were collocated instead of staggered thereby susceptible to "checkerboard" instability [17, 25].

2. In our earlier work [20], density, temperature, volume fraction, and centroid were discretized for each material, separately. The velocity in this previous work was a single variable that represented the mass weighted average of each materials' velocity. In the present work, density, volume fraction, and centroid are discretized for each material separately. The velocity is still a mass weighted average of each materials' velocity. The temperature is also a mass weighted average of each materials' temperature. The rationale for having the temperature as a mass weighted average is that we observed temperature overheating or undercooling in computational cells with "small" volume fraction.

4 temporal discretization

0. Initially the fluid (Newtonian and/or Non-Newtonian) volume fractions tessellate the whole computational domain and the elastic materials (in which we distinguish from Non-Newtonian fluids in which the characteristic wave speed $\sqrt{\tilde{\eta}_P/\rho}$ (see (33)) or $\sqrt{G/\rho}$ (see (34)) has the same order of magnitude, or larger, as the sound speed of the material) tessellate the "elastic material domain." See figure **??**. For stability purposes, the velocity is discretized at face centroids and the configuration tensor off-diagonals are discretized at edge centers[37, 17, 25] (see Figure **??**).

1. for $k = 0, 1$

2. cell integrated semi-Lagrangian advection using $\vec{u}^{(k)}$. See section **??**.

3. extrapolate $\vec{u}^{(k)}$ from the elastic regions into the fluid regions: $\vec{u}^{extrap}$

4. advect the elastic material volume fractions and centroids using $\vec{u}^{extrap}$.

5. If phase change is modeled, see section **??**

6. Jaumann derivative equation or upper convective derivative. See section **??**.

7. for $r = 0, 1$

8. Viscosity; elastic materials clamped if $r = 1$. See section **??**.

$$\frac{\vec{u}^{*,k,r} - \vec{u}^{CISL,k}}{\Delta t} = F_{viscosity}(\vec{u}^{*,k,r}) \tag{63}$$

9. Thermal diffusion and species diffusion. See section **??**.

$$\frac{\vec{T}^{*,k,r} - \vec{T}^{CISL,k}}{\Delta t} = F_{heat}(T^{*,k,r}) \tag{64}$$

10. Elastic forces ($r = 0$); see section **??**.

$$\frac{\vec{u}^{**,k} - \vec{u}^{*,k,r}}{\Delta t} = F_{elastic} \tag{65}$$

11. surface tension forces ($r = 1$); see section **??**.

$$\frac{\vec{u}^{**,k} - \vec{u}^{*,k,r}}{\Delta t} = F_{elastic} \tag{66}$$

12. Pressure gradient; elastic materials clamped if $r = 1$. See section **??**:

$$\vec{u}^{**,k} = \vec{u}^{(k+1)} + \Delta t \frac{\nabla p^{(k+1)}}{\rho} \tag{67}$$

$$\frac{p^{(k+1)} - p(\rho, e)}{\Delta t} = -\Delta t \rho c^2 \nabla \cdot \vec{u}^{(k+1)} \tag{68}$$

12. end for $r = 0, 1$

13. end for $k = 0, 1$

Remarks:

▷ steps 3 and 4 above are necessary to allow slip between fluid and elastic materials without unphysically modifying the elastic boundary. See sections **??** and **??** which demonstrate the effectiveness of this algorithm.

- ▷ steps 3 and 4 can induce a violation of mass, momentum, and energy conservation for the non-elastic materials at the elastic fluid boundaries, but we have found the error to be minimal and converge to zero with grid refinement (see sections **??** and **??**).
- ▷ similar ideas, as steps 3 and 4, were implemented in [50].
- ▷ the outer for loop, $k = 0, 1$, was implemented in [20]. THIS OUTER LOOP MIGHT BE OPTIONAL SINCE (A) VELOCITY ON MAC GRID AND (B) A SINGLE VARIABLE IS USED FOR ENERGY IN CONTRAST TO A SEPARATE ENERGY VARIABLE FOR EACH MATERIAL.
- ▷ the inner loop, $r = 0, 1$, was recommended in the article by [36].
- ▷ The effect of surface tension forces on the motion of elastic materials is not considered; so surface tension is not included when $r = 0$.
- ▷ The pressure and velocity is tightly coupled between all materials; regardless of elastic, fluid, or rigid. See step 11 above in our algorithm also see [37, 36]. Thereby we our method avoids the stability problems discussed in [4] and [3].
- ▷ In contrast to [20], which is a collocated method, our method is a staggered grid (conservative) method. In a collocated method, for incompressible flow, it is possible for the discretized projection step, even for constant density case,

$$U = (I - grad(divgrad)^{-1}div)V \tag{69}$$

  to be non-unique[17, 25]. Also in contrast to [20], our method preserves overall conservation (of non elastic fluids) of energy in contrast to preserving conservation of energy overall and for each individual material. This simplification was necessary in order to avoid temperature overheating at material boundaries with volume fractions close to zero or one.

5 phase change algorithm

See the following articles and below: [41], [40], [48].

1. For each cell in a narrow band about the interface, use the closest point map to find the corresponding closest point on the interface. Approximate the phase change speed, $|\dot{m}|/\rho$ at the closest point.

2. extend the phase change speed $|\dot{m}|/\rho$ further into bulk regions.

3. If distributing the mass source term due to expansion or compression, $\dot{m}$, to the destination material side, then the velocity on the interface corresponds to the source material velocity and the overall interface velocity is:

$$\vec{u}_{source} + \frac{|\dot{m}|}{\rho_s}\vec{n}_s \tag{70}$$

If distributing $\dot{m}$ to the source material side, then the velocity on the interface corresponds to the destination material velocity and the interface velocity is:

$$\vec{u}_{dest} + \frac{|\dot{m}|}{\rho_d}\vec{n}_s \tag{71}$$

4. Using step 3 as a guide, do unsplit CISL phase change advection. If distributing $\dot{m}$ to the destination material side, the phase change contribution to the interface velocity is

$$\frac{|\dot{m}|}{\rho_s}\vec{n}_s \tag{72}$$

5. The difference in volume fraction from step 4 gives the $\dot{m}^{conservative}$ expansion source term to be applied during the pressure update step.

6. The $\dot{m}^{conservative}$ terms are distributed to one side of the interface. If distributed to the destination material side, then distribute where $F_{dest} > 1/2$. This guarantees, for linear interfaces, that all faces with at least one adjoining source material majority cell will have the source material velocity defined at that face. Please refer to Figure 1.

For compressible flow, with a predictor and a corrector step, there is a problem. For the evaporating droplet problem, the $\dot{m}$ terms are distributed to the vapor (destination). Please refer to Figure 1. If the $\dot{m}$ terms are not distributed deep enough into the bulk of the destination material, then:

1. During the predictor step, there is no problem because the advection velocity at time t^n is used in order to advect the interface that is initially defined at $t = t^n$ (Γ^n).

2. During the corrector step, there is a problem because the advection velocity at time t^{n+1} is used in order to advect the interface that is initially defined at $t = t^n$ (Γ^n). The problem is that some of the $\dot{m}$ terms distributed to the vapor side of the Γ^{n+1} interface will be on the liquid side of the $t = t^n$ (Γ^n) interface. See the blue region of Figure 1.

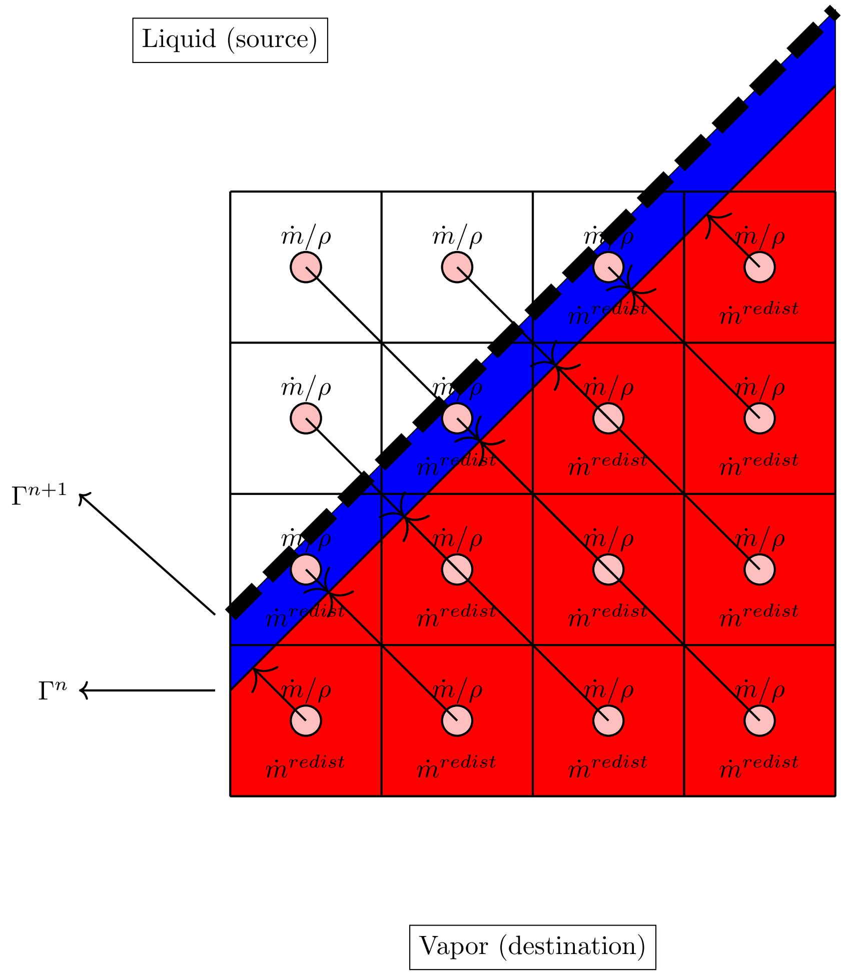

Figure 1: The phase change algorithm: (1) the phase change speed, $\dot{m}/\rho$, is approximated at all narrow band cells by evaluating $\dot{m}/\rho$ at the corresponding closest point on the interface (Γ^n). (2) The change in volume fraction due to unsplit CISL advection determines the mass source, $\dot{m}$, for the projection step. (3) the mass source, $\dot{m}$ is distributed to the vapor side of the Γ^{n+1} interface. At the next time step, the interface is advected using $\vec{u}^{source}$. For compressible flow, there is a problem if $\dot{m}$ is not distributed deep enough into the vapor material. This is because the corrector advection of Γ^n will now have some redistributed $\dot{m}$ terms on the liquid side of the Γ_n interface (the blue region).

6 Applications

1. Jet in cross flow problem (military jet engines, Pratt and Whitney)

2. Diesel injectors

3. Rotating Detonation Engines (see latest article with Everett Wenzel and Marco Arienti).

7 Algorithm

1. Cell integrated semi Lagrangian advection

2. Cell integrated semi Lagrangian advection due to phase change.

3. Viscosity, Viscoelastic, and elastic forces.

4. Temperature Diffusion

5. Mass Fraction Diffusion

6. Pressure Gradient

7. Go back to step 1.

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